3	**(a)**		$(x+2)(x+3) \quad (x+2) \quad (x+3)$ $1 = A(x+3) + B(x+2)$ $x = -3 \Rightarrow 1 = -B \Rightarrow B = -1$ $x = -2 \Rightarrow 1 = A$ So $\dfrac{1}{(x+2)(x+3)} = \dfrac{1}{(x+2)} - \dfrac{1}{(x+3)}$	**M1** **M1** **A1** **[3]**	**1.1a** **1.1** **1.1**	answer. For clearing the fractions For one appropriate substitution For correct completion	
3	**(b)**		$\int \left(\dfrac{1}{(x+2)} - \dfrac{1}{(x+3)} \right) \mathrm{d}x = \ln\lvert x+2 \rvert - \ln\lvert x+3 \rvert \ [+c]$ $\ln \left\lvert \dfrac{x+2}{x+3} \right\rvert + c$	**M2** **A1**	**1.1a** **1.1** **2.2a**	For both terms correct FT their fractions **M1** for one of their terms correct **A0** if modulus sign missing or if further work eg to try and find c	Need brackets or better (eg single fraction Ignore ' $f(x) =$ '

Question			Answer	Marks	AOs	Guidance	
				[3]			